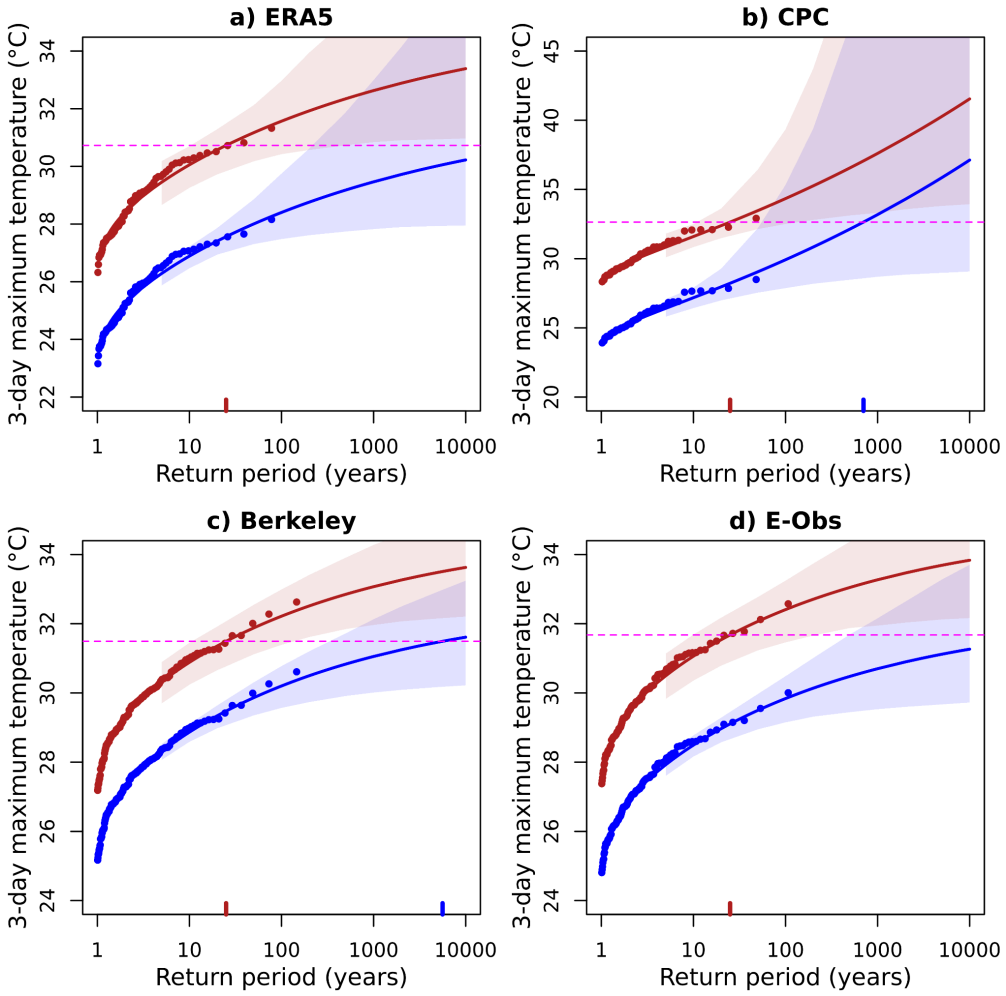


Figure 7: Statistical fits to Tx3x, in a) ERA5, b) CPC, c) Berkeley, and d) E-Obs. The influence of the current GMST is shown with the red vs blue probability curves, showing the present (2026) vs past 1976 warming levels. The magnitude of the 1-in-25-year event is highlighted with a horizontal purple line.

Index	Month(s)	Return period used for analysis	Change since 2003 (~0.6C warming)		Change since 1976 (~1.1C warming)	
			Probability ratio	Change in magnitude (C)	Probability ratio	Change in magnitude (C)
Tx3x	June	100	286 (4.32, >1m)	1.90 (0.93, 3.10)	>1m (11.9, >1m)	3.45 (1.71, 5.69)
	Annual	25	13.9 (2.28, 4980)	1.66 (0.82, 2.84)	580 (3.62, >1m)	3.04 (1.49, 5.21)
Tn3x	June	100	295 (2.88, >1m)	1.31 (0.78, 1.84)	>1m (6.75, >1m)	2.40 (1.42, 3.37)
	Annual	20	163 (4.04, >1m)	0.96 (0.31, 1.66)	>1m (18.3, >1m)	1.75 (0.57, 3.04)